Reasoning Failures in LLMs: Evaluating Robustness on Mathematical Word Problem (MWP) Variations

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Section 1/2

1. Introduction

Large Language Models (LLMs) have shown strong performance in solving mathematical word problems (MWPs). However, recent studies suggest that their reasoning is not always stable. Even small changes in a problem, such as wording, structure, or additional information, can lead to incorrect answers. This project evaluates how robust LLM reasoning is on mathematical word problems by testing how model performance changes when the same problem is modified in different ways.

2. Paper Reference

This study is based on the GSM-Symbolic framework proposed by Gulati et al. (2024), which evaluates the reasoning ability of large language models (LLMs) on MWP when small changes are made. In this framework, symbolic variations involve changing only the numbers in a problem while keeping the logic the same. No Operation (No-Op) variations involve adding extra sentences that are not relevant to the solution. Increased complexity refers to making the problem harder by adding more steps, conditions, or constraints. These variations help evaluate whether LLMs truly understand the problem or rely mainly on patterns.

3. Objective

The following objectives were defined to guide the evaluation of LLM reasoning robustness:

- To evaluate the performance of LLMs on a baseline mathematical word problem.
- To assess robustness under symbolic variations, involving changes in numerical values.

- To analyze the impact of No-Op variations, where irrelevant information is added without affecting the core logic.
- To examine model performance under increased complexity, including additional clauses and multi-step reasoning.
- To identify common reasoning failures and inconsistencies across different LLMs

4. Methodology

This study adopts a variation-based evaluation approach inspired by the GSM-Symbolic framework, focusing on mathematical word problems.

The methodology followed is as follows:

- **Baseline Selection:** A base mathematical word problem was selected.
- **Symbolic Variations:** Numerical values were modified while keeping the logic unchanged to test generalization.
- **No-Op Variations:** Irrelevant sentences were added to evaluate whether models can ignore unnecessary information.
- **Complexity Variations:** Additional clauses, multi-step reasoning, and real-world constraints were introduced to increase problem difficulty.
- **Extension Beyond GSM:** More complex variations were included to better evaluate modern LLMs and identify robustness failures.
- **Evaluation:** All variations were tested across multiple LLMs, and model performance was evaluated based on the correctness of responses.

All problem sets, prompts, and recorded LLM responses used in this study are made available on the project GitHub repository at: https://github.com/60106908/LLM-robustness-testing.git

5. Problem Set

The following variations were created based on a baseline mathematical word problem to evaluate reasoning robustness under different conditions. The table below summarizes the different problem variations used in this study.

Prompt ID	Variation Type
V1	Baseline (original GSM problem)
V2	GSM Symbolic (decimal values)

V3	GSM Symbolic (large numerical values)
V4	Increased complexity (additional constraints)
V5	GSM NoOp (irrelevant sentence added)
V6	Increased complexity (additional clauses)
V7	Increased complexity (multi-step reasoning)
V8	Increased complexity (algebraic transformation)
V9	Increased complexity (logical constraints)
V10	GSM No-Op + complexity (multiple irrelevant + multi-step)

Table 1: Summary of problem variations used in the study

The variations used in this study follow the GSM-Symbolic framework, where the baseline problem was first modified using symbolic changes such as altering numerical values. Building on this, additional variations were created by increasing the complexity of the problems through the introduction of extra clauses, multi-step reasoning, and real-world constraints. Furthermore, No-Op perturbations were applied by adding irrelevant sentences that do not affect the final answer. While this approach is based on the methodology proposed in the GSM-Symbolic paper, this study extends it by introducing more complex variations to evaluate reasoning robustness in modern LLMs.

The full problem statements are provided in Appendix A.

6. Results

All prompts used in this evaluation are provided in Appendix A. The compiled results obtained from multiple LLMs are presented below.

The performance of different LLMs was evaluated using the problem variations described above. The following figures and tables present the performance of different LLMs across all problem variations, including a summary matrix and a comparison of overall model accuracy.

Prompt ID	Variation Type	Expected Answer	ChatGPT	Gemini	DeepSeek	Claude	Error Type
V1	Baseline (original GSM problem)	14	14	14	14	14	No Error

V2	GSM Symbolic (decimal values)	13	13	13	13	13	No Error
V3	GSM Symbolic (large numerical values)	305239	305239	305239	305239	305239	No Error
V4	Increased complexity (additional constraints)	242157	242157	242157	242157	242157	No Error
V5	GSM NoOp (irrelevant sentence added)	498	498	498	498	498	No Error
V6	Increased complexity (additional clauses)	6345	6826	6345	6345	6345	logic/conceptual error
V7	Increased complexity (multi-step reasoning)	32.46	32.46	32.46	45.74	32.46	Wrong base rate applied.
V8	Increased complexity (algebraic transformation)	Ahmed=140, Sara=70, Khalid=55	Ahmed=39, Sara=19.5, Khalid=4.5	Ahmed=140, Sara=70, Khalid=55	Ahmed=39, Sara=19.5, Khalid=4.5	Ahmed=140, Sara=70, Khalid=59	Misidentification of subject
V9	Increased complexity (logical constraints)	$178.50	$178.51	$178.52	$178.53	$178.54	No Error
V10	GSM NoOp + complexity (multiple irrelevant + multi-step)	118129840	4448805	123482935	4448805	118129840	used irrelevant info for calculating, tried multiple times these three models keeps giving different answers

Table 2: Performance of LLMs across different problem variations

LLM	***V1***	***V2***	***V3***	***V4***	***V5***	***V6***	***V7***	***V8***	***V9***	***V10***
ChatGPT	✓	✓	✓	✓	✓	x	✓	x	✓	x
Gemini	✓	✓	✓	✓	✓	✓	✓	✓	✓	x
DeepSeek	✓	✓	✓	✓	✓	✓	x	x	✓	x
Claude	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 3: Summary matrix of model performance (✓indicates correct response)

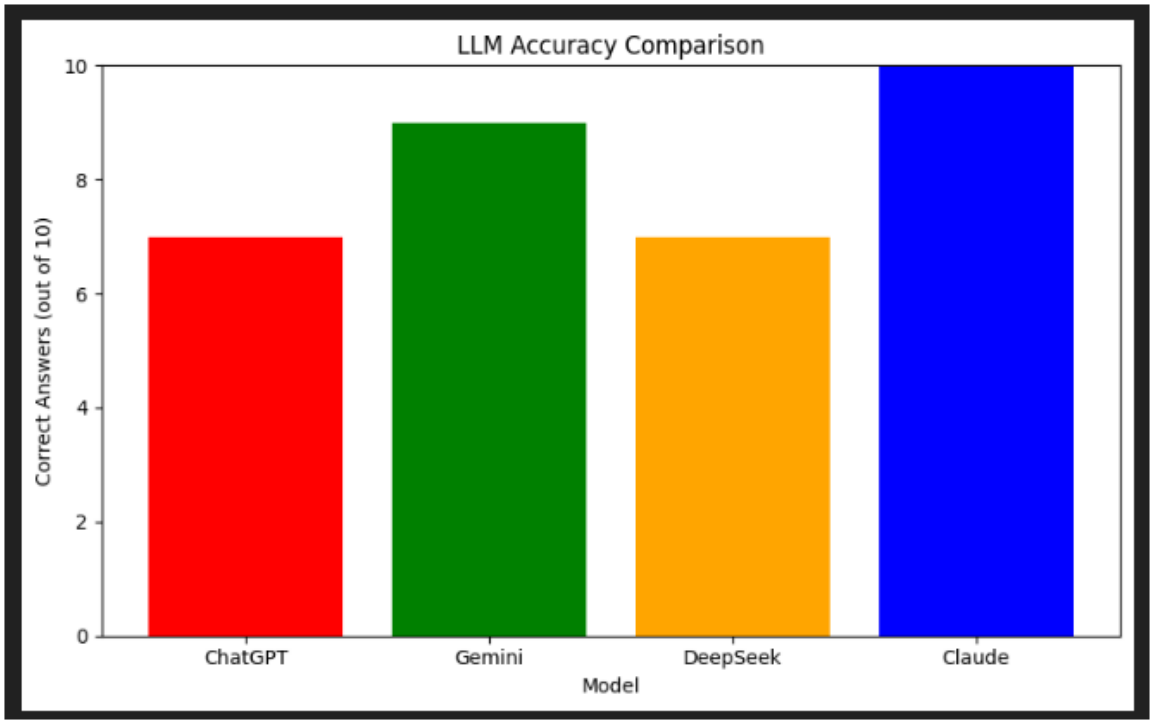

Figure 1 below presents a comparison of overall model accuracy across all variations

Claude achieved the highest accuracy (100%), followed by Gemini (90%), while ChatGPT and DeepSeek showed lower performance (70%), particularly under complex and NoOp variations.

7. Analysis

The following analysis highlights key observations and patterns in model performance across different problem variations

1. Strong performance on baseline and symbolic variations

All models performed consistently well on the baseline (V1) and symbolic variations (V2, V3). This indicates that LLMs are capable of handling direct numerical substitutions and pattern-based reasoning effectively. Even with large numerical values, the models maintained high accuracy, suggesting robustness in arithmetic scaling.

2. Increased errors with structural complexity

As the complexity of the problems increased, performance began to degrade. In V6, ChatGPT produced an incorrect answer, indicating a logic or conceptual error when handling additional clauses. This suggests that multi-step reasoning introduces instability in maintaining intermediate steps.

3. Sensitivity to multi-step reasoning

In V7, DeepSeek produced an incorrect result due to applying an incorrect base rate. This highlights that models may misinterpret key constraints when multiple rules are involved, leading to reasoning errors rather than simple arithmetic mistakes.

4. Difficulty in algebraic reasoning

In V8, some models failed to correctly identify relationships between variables, producing incorrect outputs. This indicates limitations in handling algebraic transformations and symbolic relationships compared to straightforward arithmetic tasks.

5. Minor inconsistencies in logical constraint problems

In V9, models produced slightly different answers despite being close to the expected value. This suggests that LLMs can approximate logical reasoning but may introduce small inconsistencies when combining multiple constraints.

6. Significant failure under No-Op and high complexity

The most notable performance drop was observed in V10, where multiple models produced incorrect and inconsistent answers. This variation combined irrelevant information with multi-step reasoning, leading to confusion. This demonstrates that LLMs are highly sensitive to irrelevant information (No-Op) when combined with increased complexity.

7. Overall model performance comparison

Among the models evaluated, Claude demonstrated the most consistent performance across all problem variations, correctly solving even complex and combined scenarios. Gemini performed well but showed occasional errors in more complex tasks. ChatGPT and DeepSeek showed relatively lower performance, particularly when handling multi-step reasoning and problems involving irrelevant information. This suggests that while all models perform well on simpler tasks, their robustness varies significantly under more challenging conditions.

8. Conclusion

This study tested how well different LLMs can solve problems when the questions are slightly changed using GSM-style variations, added complexity, and irrelevant information. The results show that LLMs perform well on simple and direct problems, especially when only numbers are changed. However, their performance decreases when the problems become more complex. The models had difficulty with multi-step reasoning, algebra-based problems, and questions that included irrelevant information. The biggest errors happened when both complexity and irrelevant information were combined. These findings clearly show that LLM reasoning is not robust. Small changes in the problem, even when the core logic remains the same, can lead to incorrect answers. This demonstrates

clear failures in LLMs. Overall, the results suggest that LLMs do not always truly "reason" through problems. Instead, they often rely on patterns, making them less reliable when problems become more complex or confusing.

9. References

Gulati, A., et al. (2024). GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models. arXiv preprint arXiv:2410.05229.

https://arxiv.org/abs/2410.05229

Appendix A: Prompts

ID	Variation Type	Prompt
V1	Baseline (original GSM problem)	Sophie watches her nephew. The bag of building blocks has 31 blocks in it. The bin of stuffed animals has 8 stuffed animals inside. The tower of stacking rings has 9 multicolored rings on it. Sophie recently bought a tube of bouncy balls, bringing her total number of toys for her nephew up to 62. How many bouncy balls came in the tube?
V2	GSM Symbolic (decimal values)	Priya watches her nephew. The bag of building blocks has 31.5 blocks in it. The bin of stuffed animals has 8.25 stuffed animals inside. The tower of stacking rings has 9.75 rings on it. Priya recently bought a tube of bouncy balls, bringing her total number of toys for her nephew up to 62.5. How many bouncy balls came in the tube?
V3	GSM Symbolic (large numerical values)	Zhang watches her granddaughter. The bag of building blocks has 1,847,293 blocks in it. The bin of stuffed animals has 938,475 stuffed animals inside. The tower of stacking rings has 756,284 rings on it. Zhang recently bought a tube of bouncy balls, bringing her total number of toys for her granddaughter up to 3,847,291.How many bouncy balls came in the tube?
V4	Increased complexity (additional constraints)	Valentina manages a toy warehouse for her 3 grandchildren. For grandchild A, she has 847,293 building blocks but 12.5% are damaged and unusable. For grandchild B, she has 293,847 stuffed animals but 3/8 are reserved for donation. For grandchild C, she has 156,729 stacking rings but every 4th ring is defective. Valentina recently received a shipment of bouncy balls for all three grandchildren combined. The total number of usable toys

		across all three grandchildren, including the new bouncy balls, is 1,284,739. How many bouncy balls did Valentina receive? If the answer results in a decimal, round up if the decimal is equal to or greater than .5, and round down if it is less than .5.
V5	GSM No-Op (irrelevant sentence added)	Bob picks 44 kiwis on Friday. Then he picks 58 kiwis on Saturday. On Sunday, he picks triple the number of kiwis he did on Friday, and on Monday, he picked double the number of kiwis he did on Sunday and Tuesday he realised 15 of them were a bit smaller than average. How many kiwis does Bob have?
V6	Increased complexity (additional clauses)	A warehouse manager is doing inventory. Shelf A contains 3,847 red boxes. Shelf B contains 2,193 blue boxes. Shelf C contains 1,756 green boxes. Shelf D contains 4,291 yellow boxes. The manager knows the total number of boxes across all shelves including a fifth shelf is 18,432. Shelf E contains some white boxes. However, 12.5% of the red boxes are damaged and must be removed from inventory. How many white boxes are on shelf E after the damaged red boxes are removed from the total?
V7	Increased complexity (multi-step reasoning)	A smart parking system uses the following pricing rules: - Base rate: $4.00 per 20-minute interval for first 60 minutes - After 60 minutes: $2.50 per 20-minute interval - After 3 hours: $1.00 per 20-minute interval - After 5 hours: $0.50 per 20-minute interval - Peak hours (8am-10am, 5pm-7pm): double the base rate for that interval - Off-peak discount: 25% off any interval between 11pm and 6am - Weekend rate: flat $18 for up to 8 hours regardless of other rates - Public holiday: 75% of weekend rate - Loyalty gold members: first 40 minutes free + 15% off final bill - Loyalty silver members: first 20 minutes free + 10% off final bill - If total before member discount exceeds $30: additional 20% off before member discount is applied - Government tax: 8.5% on final amount - Daily maximum (non-weekend): $45 before tax You are a loyalty gold member. You enter the parking garage on a regular Wednesday at 7:00am and leave at 2:30pm. Calculate the total amount you pay including tax.
V8	Increased complexity (algebraic transformation)	Ahmed has twice as many books as Sara. Sara has 15 more books than Khalid. Khalid gives Ahmed 8 books. Now he has 47 books. How many books did Ahmed, Sara and Khalid each have originally before the transfer?

V9	Increased complexity (logical constraints)	Five friends — Ali, Bana, Carlos, Dana and Emma — each bought a different number of items at a market. The number of items bought were 3, 5, 7, 11 and 13 (in some order).Use these clues to find who bought what, then answer the question below: Clue 1: Ali bought more items than Bana but fewer than Carlos. Clue 2: The difference between Dana's and Emma's items is 6. Clue 3: Carlos did not buy the most items overall. Clue 4: Bana bought a prime number of items and so did Emma. Clue 5: Dana bought fewer items than Bana. After buying their items, each person spent money as follows: Items costing under 10 units: $3.50 each- Items costing 10 or more units: $5.25 each. What is the total amount spent by all five friends combined?
V10	GSM No-Op + complexity (multiple irrelevant + multi-step)	Hassan watches his granddaughter. The bag of building blocks has 847395 blocks in it, but 5 of them were small. The bin of stuffed animals has 2938475 stuffed animals inside; however, 300 of them were torn. The tower of stacking rings has 1567225 rings on it and 10000 were broken. Hassan recently bought a tube of bouncy balls, bringing his total number of toys for his granddaughter up to 123482935 but he like only 1482935 so he ignore other counts. And then he threw away 2/3 of the bouncing ball. How many bouncy balls actually came in the tube?